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Activity

① Medical image Enhancement

② Which gray-level transformation would you recommend?

③ Contrast stretching / Log Transformation.

④ Why is it suitable?

⑤ It increases the contrast b/w bones and soft tissues, making darker regions more visible.

⑥ What effect will it have on the image?

⑦ Bones remain clear, and dark tissues become brighter and easier to analyze.

II. Night-Time Surveillance.

① Which Transformation would improve the visibility?

② Log Transformation

③ Why is it more suitable than negative transformation?

④ Log transformation enhances dark pixel values, while negative transformation only inverts brightness without improving visibility.

⑤ Predict the appearance of the enhanced image?

⑥ The parking lot will appear brighter, with vehicles and objects becoming more visible.

III. Fingerprint Recognition

① Which gray-level transformation would you apply?

① Contrast stretching

② Explain why

① It increases the contrast b/w fingerprint ridges and valleys, making the fine details clearer for accurate recognition.

③ What would happen if a log transformation were applied instead?

① The image would become brighter, but ridge contrast may not improve significantly, making fingerprint details less distinct than with contrast stretching.

